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Network Functions Virtualisation (NFV);
Testing;
Test Case Description Template Specification

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Foreword

This Group Specification (GS) has been produced by ETSI Industry Specification Group (ISG) Network Functions Virtualisation (NFV).

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

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1 Scope

The present document is based on the requirement of establishing a test case description convention between operators and providers in DevOps automated testing. It proposes a test case description template, to be used for standardizing the input and output information exchanged for test execution and result analysis.

The standardized test case description template in the present document aims at determining a standardized machine-readable format to for (but not restricted to):

1) Test case description information
2) Test case input information
3) Test case script designation information
4) Test case output information

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI GS NFV-TST 010: "Network Function Virtualisation (NFV) Release 2; Testing; API Conformance Testing Specification".

[2] ETSI GS NFV-SOL 001: "Network Functions Virtualisation (NFV) Release 3; Protocols and Data Models; NFV descriptors based on TOSCA specification".

[3] ETSI GS NFV-SOL 004: "Network Functions Virtualisation (NFV) Release 2; Protocols and Data Models; VNF Package and PNFD Archive specification".

[4] ETSI GS NFV-SOL 006: "Network Functions Virtualisation (NFV) Release 3; Protocols and Data Models; NFV descriptors based on YANG Specification".

[5] ETSI GS NFV-SOL 007: "Network Functions Virtualisation (NFV) Release 3; Protocols and Data Models; Network Service Descriptor File Structure Specification".

[6] ETSI GS NFV-TST 009: "Network Functions Virtualisation (NFV) Release 3; Testing; Specification of Networking Benchmarks and Measurement Methods for NFVI".

[7] ETSI GS NFV-SOL 014: "Network Functions Virtualisation (NFV) Release 3; Protocols and Data Models; YAML data model specification for descriptor-based virtualised resource management".

[8] "YAML Ain't Markup Language (YAML™) Version 1.2", 3rd Edition. Oren Ben-Kiki, Clark Evans, Ingy döt Net.

NOTE: Available at http://www.yaml.org/spec/1.2/spec.html.

[9] IETF RFC 8259: "The JavaScript Object Notation (JSON) Data Interchange Format".

NOTE: Available at https://tools.ietf.org/html/rfc8259.

[10] JSON Schema.

NOTE: Available at https://json-schema.org/.

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long-term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

Not applicable.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purpose of the present document, the following terms apply:

configuration: element of test case to describe information like network topology and the specific structure of the test environment

pre-test conditions: element of test case to describe the dependent conditions before the test sequence is performed in which the SUT/Test Function should be ready for executing the test and in which the test related parameters/measurements are defined

reference function: reference implementation of NFV functional components in the test environment

NOTE: The NFV functional components which are neither SUT nor Test Function in the test environment are Reference Function, the actual entity of Reference Function depends on the specific SUT.

test environment: environment which provides all the functional elements needed for the testing execution on SUT, consists of test functions, reference functions

test function: entity that will be controlled (for example, by test controller) for test execution and monitored to obtain measurements for test results in the test environment

test PNF: physical instrument as test function in the test environment

test sequence: element of test case which contains a series of test steps listed in sequence to describe the operation of each step like controlling or checking the SUT/Test Function

test system: specialized tool (system) built for the purpose of testing that has the abilities including test case management and execution, control and communication with SUT during testing, observation and measurement of test result

test verdict: element of test case that is used to describe how to record the test result according to the checking step's result in test sequence. For functionality or API testing, test verdict describes how the result is deemed as passed or failed; for benchmark testing, test verdict describes which values should be recorded

test VNF: virtual instrument as test function in the test environment

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

API	Application Programming Interface
HTML	Hyper Text Markup Language
JSON	JavaScript Object Notation
MANO	Management and Orchestration
NFV	Network Function Virtualization
NFVI	Network Function Virtualization Infrastructure
NFVO	NFV Orchestrator
NS	Network Service
SUT	System Under Test
URL	Uniform Resource Locator
VIM	Virtual Infrastructure Manager
VM	Virtual Machine
VNF	Virtual Network Function
VNFM	VNF Manager
vFW	virtual Firewall
XML	Extensible Markup Language
YAML	YAML Ain't Markup Language

4 Test Case Description use-cases (informative)

4.1 General

The following use cases describe the steps involved in NFV automatic testing, the relevant information may be referred in the Test Case Description file. The use cases capture the generic processes as well as the actions required to be performed by actors playing different roles in order to identify the requirements for the standard Test Case Description format. All the use cases presented in this clause are informative.

For the purpose of the use cases, the roles identified in table 4.1-1 have been identified.

Table 4.1-1: List of roles

Role	Description
Test Case Description Provider	Provide the Test Case Description file according to corresponding automated test.
Test Case Description Consumer	System that parses the Test Case Description file and use the information obtained from the test description file to automatically execute the test. (For example, it could be a test framework.)

4.2 Test Case Description composition

Test cases are designed for SUT (in the present document are functional components of the NFV architecture), and automated tests are implemented. The standardized Test Case Description defined in the present document provides key information to describe the test, information about deployment automation, information about automated test execution, and information about test result collection. The Test Case Description composition encompasses steps to describe that the content in Test Case Description will be specified by which role.

Table 4.2-1: Roles

#	Role
1	Test Case Description Provider

Table 4.2-2: Pre-conditions

#	Pre-conditions	Comment
1	Test case has been designed; automated test has been implemented	

Table 4.2-3: Post-conditions

#	Post-conditions	Comment
1	A Test Case Description file for corresponding test case	

Table 4.2-4: Base Flow

#	Role	Action/Description
1	Test Case Description provider	According to the automated test, with the standard format, specify the information including: • key information to describe the test; • information about deployment automation; • information about automated test execution; • information about test result collection.

4.3 Test case selection

By parsing the key description information of the test case provided in the test description files, the Test Case Description Consumer may have the capability to select test cases according to the actual test task.

Table 4.3-1: Roles

#	Role
1	Test Case Description Consumer

Table 4.3-2: Pre-conditions

#	Pre-conditions	Comment
1	Test Case Description composition is done	

Table 4.3-3: Post-conditions

#	Post-conditions	Comment
1	The test cases to be executed have been selected	

Table 4.3-4: Base Flow

#	Role	Action/Description
1	Test Case Description Consumer	Obtain Test Case Description files.
2	Test Case Description Consumer	Parse the key description information of the test case.
3	Test Case Description Consumer	Select the test cases to be executed.

4.4 Automated test deployment

Automated test is closely related to the execution environment. SUT, Test Functions and Reference Functions may need to be deployed automatically before the test execution. Considering the functional components of the NFV architecture as different type of software, there are kinds of deployment automation solutions.

Typical examples of those solutions are:

- Setting up virtual machines by delivering VM images
- Managing VNF deployments via OpenStack Heat or according to ETSI GS NFV-SOL 001 [2], ETSI GS NFV-SOL 004 [3], ETSI GS NFV-SOL 006 [4] and ETSI GS NFV-SOL 007 [5]
- Managing container deployments via Kubernetes®
- Installing application software on top of an operating system

When SUT is a VNF under test, the relevant test functions are VNFs too, the deployment will be that VNFs are instantiated through NFV MANO. In this case, Test Case Description file may need to specify the resource file information required for deployment operation: VNF package, and other required parameters depending on demands.

NOTE: Other situations for non-VNF SUTs are FFS.

Table 4.4-1: Roles

#	Role
1	Test Case Description Consumer

Table 4.4-2: Pre-conditions

#	Pre-conditions	Comment
1	The test cases have been selected to be run	

Table 4.4-3: Post-conditions

#	Post-conditions	Comment
1	The SUT/Test Function/Reference Function involved in the automated test have been deployed	

Table 4.4-4: Base Flow

#	Role	Action/Description
1	Test Case Description Consumer	Obtain Test Case Description files.
2	Test Case Description Consumer	Map the related parameters to the corresponding values in actual environment.
3	Test Case Description Consumer	With the parameter values, perform the deployment operation via execute automation scripts.

4.5 Automated test execution

If the role that implements the automated test and the role that executes the automated test are not hosted by the same organization, a method is needed between these 2 roles to provide the information about how to execute automated testing.

The test case description file may contain such information, so that the Test Case Description consumer can use the information to execute the test via parsing the machine-readable test case description file.

During this process, the Test Case Description consumer may need to know:

- By which type of operation the test is performed (e.g. by running a command, by executing a test script or by call an API).
- The details about the exact operation.
- On which system the operation is done.

- The relevant files for execution (e.g. the test script file).
- The parameters related to actual environment (mapping these parameters to the corresponding values in actual environment).
- The parameters specified for values related to test steps (may have default values, but also capable to specify the customized values when execute the test).

Table 4.5-1: Roles

#	Role
1	Test Case Description Consumer

Table 4.5-2: Pre-conditions

#	Pre-conditions	Comment
1	The SUT/Test Function/Reference Function involved in the automated test have been deployed	

Table 4.5-3: Post-conditions

#	Post-conditions	Comment
1	The automated test execution has finished	

Table 4.5-4: Base Flow

#	Role	Action/Description
1	Test Case Description Consumer	Obtain the test case description files.
2	Test Case Description Consumer	Parse and confirm the exact operation to execute the test, fetch or find the related files.
3	Test Case Description Consumer	Map the related parameters to the corresponding values in the actual environment.
4	Test Case Description Consumer	Specify the values related to test steps as parameters of test execution. (Optional)
5	Test Case Description Consumer	Execute the automated test.

4.6 Test result collection

The result of automated test execution may be presented differently depending on the implementation. For example, it may be printed directly in the output, or saved as files with format like JSON, XML, HTML, text. Test results presented in different forms increase complexity in the collection of test results.

The test case description file may provide information to describe how the test results are collected, such as by saving the printed information directly, by fetching information through the API, or by obtaining files with specified path/location. Then the Test Case Description Consumer can use the information above to collect test results.

Table 4.6-1: Roles

#	Role
1	Test Case Description Consumer

Table 4.6-2: Pre-conditions

#	Pre-conditions	Comment
1	Test case description file contains information that can be parsed by Test Case Description Consumer about when to collect results (collection start/stop) and how to collect results	
2	The automated test execution is ready to begin	

Table 4.6-3: Post-conditions

#	Post-conditions	Comment
1	Test result has been collected	

Table 4.6-4: Base Flow

#	Role	Action/Description
1	Test Case Description Consumer	Parse the test description file and confirm how to collect the test result.
2	Test Case Description Consumer	Collect test results based on the information parsed from the file.

5 Requirements for Test case description template

5.1 Generic Requirements

Table 5.1-1 specifies generic requirements applicable to the standardised test case description file.

Table 5.1-1: Generic requirements for test case description file

Numbering	Requirement Description
TEST_CASE_DESC.GEN.001	The test case description file shall contain the key information to describe the test, information about deployment automation, information about automated test execution, information about test result collection.

5.2 Requirements for test case description information

Table 5.2-1 specifies requirements applicable to the test case description information.

Table 5.2-1: Requirements for test case description information

Numbering	Requirement Description
TEST_CASE_DESC.TD.001	The test description shall support a way to identify uniquely the test.
TEST_CASE_DESC.TD.002	The test description shall include the type of testing for test case selection.
TEST_CASE_DESC.TD.003	The test description shall include the information about SUT for test case selection.
TEST_CASE_DESC.TD.004	The test description shall include machine readable key information of test case for test case selection.
TEST_CASE_DESC.TD.005	The test description shall include human readable key information of test case for understanding.
TEST_CASE_DESC.TD.006	The test description shall include the information to describe the pre-conditions and post-conditions of test case.
TEST_CASE_DESC.TD.007	The test description shall include the information to describe main steps of test case.

5.3 Requirements for test case inputs information

Table 5.3-1 specifies requirements applicable to the test case inputs information.

Table 5.3-1: Requirements for test case inputs information

Numbering	Requirement Description
TEST_CASE_DESC.Inputs.001	The inputs shall provide the required information for the execution of test case.
TEST_CASE_DESC.Inputs.002	The inputs shall provide human readable description of the information for the test case execution, including functional description, default values, etc.

5.4 Requirements for test case script execution information

Table 5.4-1 specifies requirements applicable to the test case script execution information.

Table 5.4-1: Requirements for test case script execution information

Numbering	Requirement Description
TEST_CASE_DESC.SE.001	The test case description shall identify the location and the scripting language of the script to be executed.
TEST_CASE_DESC.SE.002	The test case description shall define the environment requirements of script execution.

5.5 Requirements for test case outputs information

Table 5.5-1 specifies requirements applicable to the test case outputs information.

Table 5.5-1: Requirements for test case outputs information

Numbering	Requirement Description
TEST_CASE_DESC.Outputs.001	If required, the output shall include the result of functional tests.
TEST_CASE_DESC.Outputs.002	If required, the output shall include one or more measurements of test result.
TEST_CASE_DESC.Outputs.003	The output shall be available in a machine readable format.
TEST_CASE_DESC.Outputs.004	The output shall be available in a human readable format.

6 Test case description information model

6.1 Introduction

The clauses below define the information elements related to the Test Case Descriptor, which is also the Test case description template.

6.2 TestCaseDescriptor information element

6.2.1 Introduction

A TestCaseDescriptor is a test case description template which describes a test case in terms of automated test including inputs, execution, outputs.

6.2.2 Attributes

The attributes of the TestCaseDescriptor information element shall follow the indications provided in table 6.2.2-1.

Table 6.2.2-1: Attributes of the TestCaseDescriptor information element

Attribute	Qualifier	Cardinality	Content	Description
id	M	1	Identifier	Specifies this template uniquely.
testDescription	M	1	TestDescription	Specifies a test description plus the type of test, type of evaluation, SUT to be tested, provider of test case. See note. See clause 6.3.
input	M	0..N	TestInput	Specifies the input information of test case. See clause 6.4.
scriptExecution	M	1	ScriptExecution	Specifies the script designation information of test case. See clause 6.5.
output	M	0..N	RequiredTestOutput	Specifies the output information of test case. See clause 6.6.
NOTE: The concept of a test description is coincident with the test description defined in other TST specifications like ETSI GS NFV-TST 010 [1].				

6.3 TestDescription information element

6.3.1 Introduction

A TestDescription information element describes the identifier of test case, the objective of test case, the type of testing, the information about SUT, the configuration of test case, the provider of test case, the pre-conditions, the applicability, the post-conditions and the test sequence of test case, etc.

6.3.2 Attributes

The attributes of the TestDescription information element shall follow the indications provided in table 6.3.2-1.

Table 6.3.2-1: Attributes of the TestDescription information element

Attribute	Qualifier	Cardinality	Content	Description
testId	M	1	String	Identifier of this test case.
testObjective	M	1	String	Provides human-readable information on what is intended to be tested in the test case.
reference	M	1	String	Indicates the specifications or requirements documents where the tested requirements are expressed.
testType	M	1	Enum	Specifies the type of testing that the test case belongs to. Allowed values: • Compliance • Functionality • Benchmarking • Interoperability
testEvaluation	M	1	Enum	Specifies how the test case evaluate the test result. Allowed values: • Verification • Measurement See note 1.
sutInfo	M	1	SutInfo	Information that identifies the SUT targeted by the test case, and properties related to this SUT. See clause 6.3.3.
provider	M	1	String	The provider of this test case. This attribute may include name of the organization (like a company name) and specific person or department that defined the test case. See note 2.
preConditions	M	0..1	PreConditions	Defines the conditions that in which states the SUT should be before undergoing the actual test execution. This element may include identifiers (testId) of other test cases which need to be executed to reach the pre-conditions. See clause 6.3.4.
configId	M	1	Identifier	The name or id to identify the Configuration for this test case. See note 3.
applicability	M	0..1	Applicability	Specifies the features which are required to be supported for this test case execution. See clause 6.3.5.
postConditions	M	0..1	String	Defines the conditions that in which states the SUT should be after the test execution.
testSequence	M	1	String	Provides human-readable brief description for the steps of test case.

NOTE 1: The 'testEvaluation' describes in which way this test case will evaluate the test result. The value can be one of two types: verification and measurement, 'verification' means at the end of the execution there will be a result as pass/fail, and 'measurement' means there will be a result value for recording or for further measure.
NOTE 2: A test case can be provided by vendor, Service Provider or third-party validator.
NOTE 3: In an ideal state, typical configurations used in specific type of testing for specific SUT are defined in relevant specifications, and name or id are assigned to those Configurations. These names or ids can be directly specified in 'configId'.

6.3.3 SutInfo information element

6.3.3.1 Introduction

The SutInfo information element includes attributes related to what SUT would be tested, the type of SUT, which version would be tested, and by which vendor the SUT is provided.

6.3.3.2 Attributes

The attributes of the SutInfo information element shall follow the indications provided in table 6.3.3.2-1.

Table 6.3.3.2-1: Attributes of the SutInfo information element

Attribute	Qualifier	Cardinality	Content	Description
sutType	M	1	Enum	Specifies the type of functional component of NFV architecture to which the SUT belongs. Allowed values: • VNF • NS • NFVO • VNFM • NFVO+VNFM • NFVI • VIM
product	M	0..1	String	Indicates what specific product to be tested in human-readable format. See note.
vendor	M	1	String	Indicates by which vendor the SUT is provided
NOTE: The value is used to identify what exactly would be tested. For example, when the SUT is a vFW VNF, the 'product' could be 'FW'.				

6.3.4 PreConditions information element

6.3.4.1 Introduction

The PreConditions information element includes attributes related to the testId values of dependent test cases, the explanation of each pre-condition and if any, the automated script to make sure pre-conditions.

6.3.4.2 Attributes

The attributes of the PreConditions information element shall follow the indications provided in table 6.3.4.2-1.

Table 6.3.4.2-1: Attributes of the PreConditions information element

Attribute	Qualifier	Cardinality	Content	Description
dependency	M	0..N	String	The testId value of other test cases which need be executed to reach the pre-conditions.
condition	M	0..N	String	Provides human-readable information on the condition that SUT should reach before test execution.
actionScript	M	0..N	String	Specifies the automated script to be run to ensure pre-conditions are achieved. See note.
NOTE: The actionScript attribute shall not be present if the operations to make sure pre-conditions is implemented as part of the test script itself.				

6.3.5 Applicability information element

6.3.5.1 Introduction

The Applicability information element includes attributes related to the functions which are required for this specific test case.

6.3.5.2 Attributes

The attributes of the Applicability information element shall follow the indications provided in table 6.3.5.2-1.

Table 6.3.5.2-1: Attributes of the PreConditions information element

Attribute	Qualifier	Cardinality	Content	Description
applicableFeature	M	0..N	String	Lists the required features to be supported in order to execute this TD.

6.4 TestInput information element

6.4.1 Introduction

The TestInput information element is used to give the input parameters and their values for enabling script execution in the template, for example, testing traffic throughput and duration. The given attributes may differ from use cases.

6.4.2 Attributes

The attributes of the TestInput information element shall follow the indications provided in table 6.4.2-1.

Table 6.4.2-1: Attributes of the TestInput information element

Attribute	Qualifier	Cardinality	Content	Description
name	M	1	String	Specifies the name of the attribute required for testing execution.
description	M	1	String	Provides human readable function description of the attribute.
type	M	1	String	Specifies the input attribute data type, it can be string, integer, etc.
value	M	1	String	Specifies the value for this attribute.

6.5 ScriptExecution information element

6.5.1 Introduction

The ScriptExecution information element is to give the information that is needed to enable the execution of the script.

6.5.2 Attributes

The attributes of the ScriptExecution information element shall follow the indications provided in table 6.5.2-1.

Table 6.5.2-1: Attributes of the ScriptExecution information element

Attribute	Qualifier	Cardinality	Content	Description
address	M	1	String	Specifies the fetching location (URL) of the script.
type	M	1	String	Specifies the scripting language type of the script, for example, python，shell script.
environmentVariable	M	0..N	keyValuePair	Specifies the parameters that are related to the environment requirements of script execution, e.g. the environment version of script execution.

6.6 RequiredTestOutput information element

6.6.1 Introduction

The RequiredTestOutput information element is to describe the information required to be returned after test execution. The information can be the results of performance metrics, which can refer to requirements number within ETSI GS NFV-TST 009 [6], or vendor-specific information as agreed with the operator.

6.6.2 Attributes

The attributes of the RequiredTestOutput information element shall follow the indications provided in table 6.6.2-1.

Table 6.6.2-1: Attributes of the RequiredTestOutput information element

Attribute	Qualifier	Cardinality	Content	Description
name	M	1	String	The name of the attribute presenting the test results (for example, requirement number in ETSI GS NFV-TST 009 [6]).
description	M	1	String	Provides human readable function description of the attribute.
type	M	1	String	The data type of the attribute, it can be string, integer, etc.

7 Test case description Data Model

7.1 Overview

The following clauses define the data model based on information model definition in clause 6, and give the YAML-based data models, referring the definition in ETSI GS NFV-SOL 014 [7].

7.2 Common data types

7.2.1 Introduction

Clause 7.2 specifies the common data types that are used for declaring the parameters and grammar elements in Test Case Description Template.

7.2.2 Simple data types

The simple data types that can be used in Test Case Description Template are defined in table 7.2.2-1.

Table 7.2.2-1: Simple data types used in Test Case Description Template

Type name	Description	Example(s)
String	A string as defined in YAML v1.2 [8].	"a string"
Number	A number as defined in IETF RFC 8259 [9] referred in JSON Schema [10].	"23", "-1.023E3"
Boolean	A data type that can take the following values: true, false. The type is defined in JSON Schema and referred in YAML v1.2 [8].	"true", "false"
NOTE: Enum and Identifier in information model are represented as string, with constrained values.		

The data model definition for the parameters that are simple data types shall comply with the following YAML syntax.

Table 7.2.2-2: Data model for a parameter

Field	Required	Description
parameter_name	yes	The name of the parameter.
type	yes	The type of the parameter.
description	yes	A human readable description for the parameter.
default	no	A default value of the parameter.
enum	no	A set of enumerated values for a parameter to restrict the possible values. It can be applicable when the parameter type is Enum in information model.

```
< parameter name >:
  type: <the type of the parameter>
  description: <the description of the parameter>
  default: <the default value of the parameter>
  enum:
    - <enumerated value 1>
    - <enumerated value 2>
       …
```

7.2.3 Structured data types

The structured data types are used for complex data types, individual structured data type is represented in the present document using ">" recursively as inline definition. There are two structured data types defined in JSON schema: `object` and `array`.

Table 7.2.3-1: Data model for {parameter name}

Parameter Name and Attributes	Type	Description
{parameter name }	{object, array}	Type of the parameter.
{description}	-	Description of the parameter.
{attribute}	{attribute type}	Type of {attribute}.
>{sub attribute}	{sub attribute type in the attribute}	Type of {sub attribute}.

The syntax of `object` for parameter definition is represented with the following definition:

```
{parameter name}:
  description: <the description of the parameter>
  type: object
  required:
       - {1st mandatory attribute}
       - {2nd mandatory attribute}
       - …
  properties:
  {1st attribute}:
         type: e.g. object
         properties:
             {sub attribute}
  {2nd attribute}:
       …
```

The syntax of `array` for parameter definition is represented with the following definition:

```
{parameter name}:
description: <the description of the parameter>
type: array
minItems: { lower bound of cardinality}
maxItems: { upper bound of cardinality}
items:
-  type: e.g. object
properties:
        {sub attribute}
```

7.3 Data model for TestDescription

7.3.1 Introduction

This clause specifies the data model for TestDescription information element which specifies the attributes needed of a specific test case to be executed.

7.3.2 TestDescription element

The parameters used when defining TestDescription shall follow the indicators provided in table 7.3.2-1.

Table 7.3.2-1: Data model for TestDescriptor

Parameter Name and Attributes	Type	Description
TestDescription	Object	Specifies a test description plus the type of test, type of evaluation, SUT to be tested, provider of test case.
>testId	String	Identifier of the test case.
>testObjective	String	Provides human-readable information on what is intended to be tested in the test case.
>reference	String	Indicates the specifications or requirements documents where the tested requirements are expressed.
>testType	String	Specifies the type of testing that the test case belongs to. Allowed values: • Compliance • Functionality • Benchmarking • Interoperability
>testEvaluation	String	Specifies how the test case evaluate the test result. Allowed values: • Verification • Measurement
>sutInfo	Object	Information that identifies the SUT targeted by the test case, and properties related to this SUT.
>>sutType	String	Specifies the type of functional component of NFV architecture to which the SUT belongs. Allowed values: • VNF • NS • NFVO • VNFM • NFVO+VNFM • NFVI • VIM
>>product	String	Indicates what specific product to be tested in human-readable format. The value is used to identify what exactly is to be tested. For example, when the SUT is a vFW VNF, the 'product' could be 'FW'.
>>vendor	String	Indicates by which vendor the SUT is provided.
>provider	String	The provider of this test case. This attribute may include name of the organization (like a company name) and specific person or department that defined the test case.
>preConditions	Object	Defines the conditions that in which states the SUT should be before undergoing the actual test execution. This attribute may include testId values of other test cases which need to be executed to reach the pre-conditions.
>>dependency	Array of String	The testId value of other test cases which need be executed to reach the pre-conditions.
>>condition	Array of String	Provides human-readable information on the condition that SUT should reach before test execution.
>>actionScript	Array of String	Specifies the automated script to be run to ensure pre-conditions are achieved.
>configId	String	The name or id to identify the configurations for this test case.
>applicability	Object	Specifies the features which are required to be supported for this test case execution.

Parameter Name and Attributes	Type	Description
>> applicableFeature	Array of String	Lists the required features to be supported in order to execute this TestDescription.
>postConditions	String	Provides human-readable information on the conditions in which states the SUT should be after the test execution.
>testSequence	String	Provides human-readable brief description for the steps of the test case.

The syntax of the TestDescription shall comply with the following definition:

```
TestDescription:
  description: >
    Specifies a test description, plus including the type of test, type of evaluation, SUT to be
tested, provider of test case and so on.
  type: object
  required:
    - testId
    - testObjective
    - reference
    - testType
    - testEvaluation
    - sutInfo
    - provider
    - configId
    - testSequence
  properties:
    testId:
      description: >
        Identifier of this test case.
      type: string
    testObjective:
      description: >
        Provides human-readable information on what is intended to be tested in the test case.
      type: string
    reference:
      description: >
        Indicates the specifications or requirements documents where the tested requirements are
expressed.
      type: string
    testType:
      description: >
        Specifies the type of testing that the test case belongs to.
      type: string
      enum:
        - Compliance
        - Functionality
        - Benchmarking
        - Interoperability
    testEvaluation:
      description: >
        Specifies how the test case evaluate the test result.
      type: string
      enum:
        - Verification
        - Measurement
    sutInfo:
      description: >
        Information that identifies the SUT targeted by the test case what SUT would be tested, and
other properties related to the SUT.
      type: object
      required:
        - sutType
        - vendor
      properties:
        sutType:
          description: >
            Specifies the type of functional component of NFV architecture to which the SUT belongs.
          type: string
          enum:
            - VNF
            - NS
            - NFVO
            - VNFM
            - NFVO+VNFM
            - NFVI
```

```
            - VIM
        product:
          description: >
            Indicates what specific product would be tested. The value is used to identify what
exactly would be tested. For example, when the SUT is a vFW VNF, the 'product' could be 'FW'
          type: string
        vendor:
          description: >
            Indicates by which vendor the SUT is provided.
          type: string
        provider:
          description: >
            The provider of this test case. This attribute may include name of the organization
(like a company name) and specific person or department that defined the test case.
          type: string
    preConditions:
      description: >
        Defines the conditions that in which states the SUT should be before undergoing the actual
test execution. This attribute may include testId values of other test cases which need to be
executed to reach the pre-conditions.
      type: object
      properties:
        dependency:
          description: >
            The testId value of other test cases which need to be executed to reach the pre-
conditions.
          type: array
          minItems: 0
          items:
            type: string
        condition:
          description: >
            Describes the condition that SUT should reach before test execution.
          type: array
          minItems: 0
          items:
            type: string
        actionScript:
          description: >
            Specifies the automated script to be run to ensure pre-conditions are achieved.
          type: array
          minItems: 0
          items:
            type: string
    configId:
      description: >
        The name or id to identify the configurations for this test case.
      type: string
    applicability:
      description: >
        Specifies the features which are required to be supported for this test case execution.
      type: object
      properties:
        applicableFeature:
          description: >
            Lists the required features to be supported in order to execute this TestDescription.
          type: array
          minItems: 0
          items:
            type: string
    postConditions:
      description: >
        Defines the conditions that in which states the SUT should be after the test execution.
      type: string
    testSequence:
      description: >
        Provides human-readable brief description for the steps of the test case.
      type: string
```

7.4 Data model for TestInput

7.4.1 Introduction

This clause specifies the data model for TestInput information element which specifies the parameters needed of a specific test case to be executed.

7.4.2 TestInput element

The parameters used when defining TestInput shall follow the indicators provided in table 7.4.2-1.

Table 7.4.2-1: Data model for TestInput

Parameter Name and Attributes	Type	Description
TestInput	Array of Object	Specifies the input parameters of test case.
> name	String	Specifies the name of the parameter required for testing execution.
> description	String	Provides human readable function description of the parameter.
> type	String	Specifies the input parameter data type, it can be string, integer, etc.
> value	String	Specifies the value for this parameter.

The syntax of the `TestInput` shall comply with the following definition:

```
TestInput:
  description: >
    Specifies the input information of test case.
  type: array
  minItems: 0 # lower bound of cardinality
  items:
   type: object
  required:
    - name
    - description
    - type
    - value
  properties:
    name:
      description: >
       Specifies the name of the parameter required for testing execution.
      type: string
    description:
      description: >
       Provides human readable function description of the parameter.
      type: string
    type:
      description: >
       Specifies the input parameter data type, it can be string, integer, etc.
      type: string
    value:
      description: >
       Specifies the value for this parameter.
      type: string
```

7.5 Data model for ScriptExecution

7.5.1 Introduction

This clause specifies the data model for ScriptExecution information element which specifies what is needed to enable the execution of the script.

7.5.2 ScriptExecution element

The parameters used when defining ScriptExecution shall follow the indicators provided in table 7.5.2-1.

Table 7.5.2-1: Data model for ScriptExecution

Parameter Name and Attributes	Type	Description
ScriptExecution	Object	Specifies the script designation information of test case.
> address	String	Specifies the fetching location (URL) of the script.
> type	String	Specifies the scripting language type of the script, for example, python, shell script.
> environmentVariable	Array of Object	List of environment variable key-value pairs that are related to the environment requirements of script execution.
>> key	String	Specify the name of the environment variable parameter.
>> value	String	Specify the value of the environment variable parameter.

The syntax of the `ScriptExecution` shall comply with the following definition:

```
ScriptExecution:
  description: >
    Specifies the script designation information of test case.
  type: object
  required:
    - address
    - type
  properties:
    address:
      description: >
       Specifies the fetching location (URL) of the script.
      type: string
    type:
      description: >
       Specifies the scripting language type of the script, for example, python, shell script.
      type: string
    environmentVariable:
      description: >
       List of environment variable key-value pairs that are related to the environment requirements
of script execution.
      type: array
      minItems: 0 # lower bound of cardinality
      items:
        type: object
        properties:
          key:
            description: >
              Specify the name of the environment variable parameter.
            type: string
          value:
            description: >
              Specify the value of the environment variable parameter.
            type: string
```

7.6 Data model for RequiredTestOutput

7.6.1 Introduction

This clause specifies the data model for RequiredTestOutput information element which specifies the information required to be returned after test execution.

7.6.2 RequiredTestOutput element

The parameters used when defining RequiredTestOutput shall follow the indicators provided in table 7.6.2-1.

Table 7.6.2-1: Data model for RequiredTestOutput

Parameter Name and Attributes	Type	Description
RequiredTestOutput	Array of Object	Specifies the output parameter of test case.
> name	String	The name of the parameter presenting the test results (for example, requirement number in ETSI GS NFV-TST 009 [6]).
> description	String	Provides human readable description of the parameter.
> type	String	The data type of the parameter, it can be string, integer, etc.

The syntax of the RequiredTestOutput shall comply with the following definition:

```
RequiredTestOutput:
  description: >
    Specifies the output parameter of test case.
  type: array
  minItems: 0 # lower bound of cardinality
  items:
   type: object
  required:
    - name
    - description
    - type
  properties:
    name:
      description: >
       The name of the parameter presenting the test results (for example, requirement number in
ETSI GS NFV-TST 009 [6]).
      type: string
    description:
      description: >
       Provides human readable description of the parameter.
      type: string
    type:
      description: >
       The data type of the parameter, it can be string, integer, etc.
      type: string
```

Annex A (informative): Examples using test description template

A.1 Example#1: VNF Traffic Forwarding Test Description

This test case is for testing VNF traffic forwarding functionality, two test VNFs will count the ingress and egress traffic frames and bits with the VNF under test to have the measurement.

The following gives an example test description of test case:

```
schema_version: 1.0
name: traffic-forward-test
description: An example of automation test for VNF traffic forwarding functionality.

Test_example:
  properties:
    id: b2CC0ce7-2222-4fC7-95ed-4330d70a2277
    TestDescription:
      testId: 45eb297c-3688-4902-9c37-88e4fd5f1e38
      testObjective: Measure the packet handling capacity of bidirectional VNF traffic forwarding
function.
      reference: clause 6.4.3.3.2 - ETSI GS NFV-SOL 002 v3.6.1
      testType: Benchmarking
      testEvaluation: Measurement
      sutInfo:
        sutType: VNF
        product: FW
        vendor: abcTelecom
      provider: xyzIntegration
      preConditions:
        dependency: d95c9680-3c67-41ff-ffdc-1ba25ab8g990
        condition: The SUT VNF and Test VNF have been instantiated successfully.
        actionScript: file:///${EXECUTION_HOME}/script/topology.py
      configId: Config_prod_TOPOLOGY
      testSequence: Step 01. Test VNF A configures the IP and port of the SUT VNF and target Test
VNF B. Step 02. Set  the duration of Test VNF A. Step 03. Count the frames and bits sent and
received by Test VNF A. Step 04. Count the frames and bits sent and received by Test VNF B.

    TestInput:
      - name: format
        description: Output formats, supported formats such as table, csv, json, yaml.
        type: string
        value: yaml
      - name: labserver-ip
        description: The IP address of license server.
        type: string
        value: 10.10.10.1
      - name: username
        description: The username used for create session.
        type: string
        value: Admin
      - name: stcv1-mgmt-ip
        description:  The management IP address of Test VNF traffic generator west.
        type: string
        value: 10.10.10.2
      - name: stcv1-test-ip
        description: The IP address of Test VNF traffic generator west test port.
        type: string
        value: 10.10.10.3
      - name: stcv2-mgmt-ip
        description:  The management IP address of Test VNF traffic generator east.
        type: string
        value: 10.10.10.4
      - name: stcv2-test-ip
        description: The IP address of Test VNF traffic generator east test port.
        type: string
        value: 10.10.10.5
      - name: sut-left-ip
        description: The IP address of SUT left port.
        type: string
        value: 10.10.10.6
      - name: Sut-right-ip
```

```
      description: The IP address of SUT right port.
      type: string
      value: 10.10.10.7
    - name: duration-time
      description: The traffic forwarding duration in seconds.
      type: integer
      value: 100

  ScriptExecution:
    address: file:///${EXECUTION_HOME}/script/traffic-forward-test.py
    type: python

  RequiredTestOutput:
    - name: Stcv1_tx_frame
      description: Transmitted frames from Stcv1.
      type: string
    - name: Stcv1_tx_bit
      description: Transmitted bits from Stcv1.
      type: string
    - name: Stcv1_rx_frame
      description: Received frames of Stcv1.
      type: string
    - name: Stcv1_rx_bit
      description: Received bits of Stcv1.
      type: string
    - name: Stcv2_tx_frame
      description: Transmitted frames from Stcv2.
      type: string
    - name: Stcv2_tx_bit
      description: Transmitted bits from Stcv2.
      type: string
    - name: Stcv2_rx_frame
      description: Received frames of Stcv2.
      type: string
    - name: Stcv2_rx_bit
      description: Received bits of Stcv2.
      type: string
```

Annex B (informative): Bibliography

ETSI GS NFV-TST 001: "Network Functions Virtualisation (NFV); Pre-deployment Testing; Report on Validation of NFV Environments and Services".

ETSI GR NFV-TST 006 (V1.1.1): "Network Functions Virtualisation (NFV); Testing; Report on CICD and Devops".

ETSI GR NFV-TST 007 (V2.6.1): "Network Functions Virtualisation (NFV) Release 2; Testing; Guidelines on Interoperability Testing for MANO".

ETSI GR NFV-TST 011 (V1.1.1): "Network Functions Virtualisation (NFV); Testing; Test Domain and Description Language Recommendations".

Annex C (informative): Change history

Date	Version	Information about changes
Jul 28th, 2020	V0.0.1	Start drafting the skeleton of TST013
Mar 05th, 2021	V0.0.2	Updated the skeleton and added already approved contents: 1) NFVTST(20)000063r2: TST013-proposed contribution on scope in clause1 2) NFVTST(20)000062r1: TST013-proposed contribution on references in clause 2 3) NFVTST(20)000064r8: TST013-proposed contribution on test case description convention in clause3 4) NFVTST(20)000061r3: TST013–proposed contribution on automated test execution in clause 4 5) NFVTST(20)000090r7: TST013-proposed contribution on TestDescription element 6) NFVTST(21)000003r2: TST013-proposed contribution on modification of the skeleton
Oct 26th, 2021	V0.0.3	1) NFVTST(21)000030r4: TST013-proposed contribution on clause 4 2) NFVTST(21)163002r1: TST013-proposed contribution on use cases clause 4.5 3) NFVTST(21)163004r2: TST013-proposed contribution on use cases clause 4.6 4) NFVTST(21)000031r2: TST013-proposed contribution on clause 5.1-5.2 requirements 5) NFVTST(21)000067r2: TST013-proposed contribution on clause 5.3 requirements 6) NFVTST(21)000081r2: TST013-proposed contribution on clause 5.4-5.5 requirements 7) NFVTST(21)000086: TST013-Update_References__of_Adding_SOL001_006
Apr 15th, 2022	V0.0.4	1) NFVTST(21)000089r2: TST013-Correct the Cardinality of Inputs in clause 6.2.2 2) NFVTST(21)000088r3: TST013-Contribution on clause 6.4 inputs information element 3) NFVTST(21)000106r2: TST013- Proposed contribution on clause 6.5 scriptExecution information element 4) NFVTST(21)000113r5: TST013- Proposed contribution on clause 6.6 outputs information element 5) NFVTST(22)000013r2: TST013- Contribution on clause 7.1~7.2.3 common data types
Jul 08th, 2022	V0.0.5	1) NFVTST(22)000018r5_TST_013_Contribution_on_clause_7_3_data_model_for_TestCaseIn 2) NFVTST(22)000029r2_TST013-Clean_up_editor_note_in_clause_6_3_3_2 3) NFVTST(22)000030r2_TST013-Update chapter 6 to align with latest discussion 4) NFVTST(22)000036r1_TST013-Contribution on clause 7.4 Data model for Inputs 5) NFVTST(22)000042r1_TST013-Contribution on clause 7.5 Data model for ScriptExecution 6) NFVTST(22)000043r1_TST013-Contribution on clause 7.6 Data model for test case Output 7) NFVTST(22)000044r2_TST013-Contribution_on_Annex_A 8) NFVTST(22)000049r1_TST013-Clean up the Editor's notes all over the document

History

Document history		
V1.1.1	September 2022	Publication